

Modification of the Crystallization Behavior of Isotactic Polypropylene by Grafting Styrene Segments

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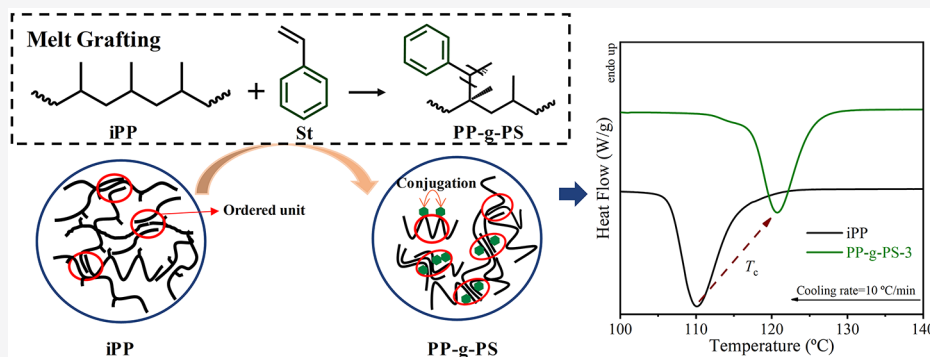


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ABSTRACT: As a typically semicrystalline polymer, the physical properties of isotactic polypropylene (iPP) are directly related to its crystallization. In this work, styrene (St) is grafted onto iPP chains by reactive processing to modify its crystallization behavior. The St grafting segments disrupt the stereoregularity of the iPP chain, leading to the appearance of a γ -crystal. More interestingly, the presence of St grafting segments can remarkably increase the crystallization temperature and crystallization rate of iPP and reduce the size of crystals. On the one hand, the rigid structure of benzene rings in St grafting segments can squeeze easily iPP chains to form the relatively ordered structure, which can be further stabilized by the mutual π - π conjugation between the benzene rings and the helix chains of iPP. On the other hand, the St grafting segments locating in the amorphous phase can hinder the growth of crystals, which leads to finer and more crystals.

INTRODUCTION

Isotactic polypropylene (iPP) is one of the most widely used commercial polymers due to its low material cost and desirable properties such as chemical resistance and easy recycling. As a semicrystalline polymer, it is well-known that the properties of iPP are strongly dependent on their crystalline structure and morphology.¹ In order to further expand the application range of iPP, it is significant to modify its crystalline behavior, e.g., crystallization rate, spherulite size, and crystal form.^{2,3} Many methods such as adding nucleating agents (NAs) and introducing special monomers have been widely investigated in both academic and industrial fields.

Generally, the purpose of introducing the NAs is to provide great amounts of nucleation sites to raise the crystallization temperature and affect the crystal morphology.⁴ The common small-molecule NAs such as sorbitol derivatives,⁵ phosphate ester salts,⁶ and aromatic amides⁷ could give iPP a superior crystallization behavior with an extremely low dosage. However, the complicated synthesis certainly boosted the cost of industrialization, affecting the markets negatively. Likewise, the novel polymeric nucleating agents, e.g., polystyrene (PS),^{8–12} thermoplastic phenolic resin,¹³ acrylonitrile–styrene

copolymer,^{8,14} and acrylonitrile–butadiene–styrene graft copolymer,^{15,16} have attracted much attention of scientists owing to their innocuity and ease of processing. Most of these polymeric NAs could refine the crystallization of iPP and induce the β -nucleation. However, their modification on the crystallization of iPP is far from satisfactory due to their poor compatibility with iPP.

Alternatively, the melt grafting of special segments onto iPP through reactive extrusion can endow iPP with unique properties intrinsically,^{17–19} despite the fact that many scientific issues remain.²⁰ Some grafted segments possess great potential for modifying the crystallization behavior of iPP.²¹ The investigation on the nonisothermal crystallization behaviors of some functionalized polypropylenes,^{22,23} e.g., polypropylene-g-acrylic acid, polypropylene-g-glycidyl methacrylate, have in-

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dictated that the introduction of polar groups, acting as a nucleating agent, leads to a faster crystallization. Simultaneously, Yu et al.²⁴ reported that the maleic anhydride grafted iPP required less energy for folding macromolecules to form nuclei. Additionally, in terms of grafted iPP, the introduction of special segments in the regular constituents of the iPP chains can intensively affect the arrangement of the iPP chain at a molecular level, which results in the crystal transition, i.e., α to γ form.^{25,26} For example, Su et al.²⁷ introduced the long chain branching via in situ polymerization of iPP and an asymmetric diene monomer, i.e., *p*-(3-butenyl) styrene. The obtained iPP with long branches crystallized from the melt as a mixture of α and γ forms. Meanwhile, the content of the γ form increased with the increase of crystallization temperature (T_c) until it reached a maximum value. However, the melt grafting of PP by a vinyl monomer such as maleic anhydride and glycidyl methacrylate causes the severe degradation of iPP and consequently a substantial deterioration in the mechanical property. To solve this problem, the addition of styrene (St) as a comonomer in the grafting system can assist in reducing the iPP chain scission and enhancing the grafting yields of monomers on iPP.^{28,29} This is because the St preferentially reacts with iPP macroradicals to form more stable styryl macroradicals, in turn promoting the grafting level.³⁰

In this work, the St was introduced solely into iPP by the melt grafting method. As expected, the degradation of iPP was dramatically inhibited. More interestingly, the introduction of a 3.4 wt % St segment can enhance greatly the crystallization temperature of iPP from 110.1 to 120.5 °C and reduce dramatically the size of crystals. This work aims to provide a detailed investigation on the crystallization behaviors of polypropylene-graft-polystyrene (PP-g-PS), e.g., crystallization temperature, crystallization kinetics, spherulite morphology, and crystal form, and to discuss the role of St grafting segments on the crystallization process.

EXPERIMENTAL SECTION

Materials. Both iPP with a commercial grade of F401 and PS with a commercial grade of 158 K were provided by Yangzi Petrochemical Company, China. The isotacticity of the iPP is more than 96%. The weight-average molecular weight (M_w), number-average molecular weight (M_n), and polydispersity index (M_w/M_n) of the iPP are 476 000, 112 000, and 4.24, respectively. Styrene (St) with a purity of 99% and dicumyl peroxide (DCP) with a purity of 98% were purchased from J&K Chemical, China. Prior to usage, St was purified by distilling under reduced pressure. The other experimental reagents including vinylcyclohexane (VCHX) with a purity of 98%, 2-vinylnaphthalene (2-VN) with a purity of 97%, xylene with a purity of 99%, and acetone with a purity of 99.5% were purchased from Sinopharm Chemical, China.

Preparation of Modified iPP by Phenyl-Contained Function Group. Modified iPPs were synthesized via the melt grafting method in the Brabender Rheomix equipped with a mixing chamber in which two sigma rotors rotated in opposite directions to ensure mixing. First, liquid monomers and peroxide were premixed with iPP in a beaker. Their compositions for the synthesis of PP-g-PS were listed in Table 1. Second, those mixed components were charged simultaneously into the mixing chamber and were reacted at 190 °C for 10 min. Finally, samples were taken from the mixing chamber. Because monomers may homopolymerize or remain unreacted, the as-synthesized samples were purified according to the

Table 1. Compositions Used for the Synthesis of PP-g-PS

samples	iPP (g)	St (mL)	DCP (g)
PP-g-PS-1	40	1	0.272
PP-g-PS-2	40	2	0.544
PP-g-PS-3	40	4	1.088

following steps: they were dissolved in hot xylene at 130 °C, precipitated in acetone, filtered, and then dried in a vacuum oven at 80 °C for 24 h.

Characterization. Some products were characterized by Fourier transform infrared spectroscopy (FTIR) of type Thermo Nicolet 5700. For that purpose, they were pressed to films of below 100 μ m thick using a hot press at 190 °C and under 20 MPa. FTIR spectra of those films were recorded with 32 scans at a resolution of 4 cm^{-1} in the region of 4000–500 cm^{-1} .

Molecular weight and distributions of iPP and modified iPP were determined by high-temperature gel permeation chromatography (GPC) of type PL-GPC220, Polymer Laboratories. Prior to measurement, the samples were dissolved in 1,2,4-trichlorobenzene at 150 °C to form a solution of 5 mg/mL. 1,2,4-Trichlorobenzene was used as the mobile phase and PS was used for the calibration.

The nonisothermal and isothermal crystallization behaviors of samples were measured using a differential scanning calorimeter (DSC) of a PerkinElmer Pyris under the nitrogen atmosphere. For the nonisothermal crystallization, samples were subjected to the following thermal cycle: they were first heated from 30 to 190 °C at a rate of 30 °C/min and isothermally maintained at 190 °C for 5 min to eliminate thermal history, then cooled to 30 °C at a rate of 10 °C/min, and finally heated again to 190 °C at a rate of 10 °C/min. The crystallinity (ϕ_m) can be calculated according to the following formula:

$$\phi_m = \frac{\Delta H}{\Delta H_0} \times 100\% \quad (1)$$

where, ΔH is the melting enthalpy, which can be obtained from the heat flow curves, and the melting of enthalpy of a perfect crystal (ΔH_0) is 207 J/g.³¹

For the isothermal crystallization, samples were first annealed at 190 °C for 5 min to eliminate the thermal history and then rapidly cooled to the desired temperature at a rate of 100 °C/min for enough time until the crystallization was fully completed. The value of relative crystallinity with time ($X(t)$) can be obtained according to the following equation:

$$X(t) = \frac{Q_t}{Q_\infty} = \frac{\int_0^t \left(\frac{dH}{dt} \right) dt}{\int_0^\infty \left(\frac{dH}{dt} \right) dt} \quad (2)$$

where Q_t is the heat released at time t during the isothermal crystallization and Q_∞ is the total heat released of the system when the crystallization process is completed.

The obtained results of relative crystallinity can be described by an Avrami model:

$$X(t) = 1 - \exp(-Kt^n) \quad (3)$$

where K is the rate constant and n is the Avrami exponent. Then, eq 3 can be transformed to

$$\log[-\ln(1 - X(t))] = \log K + n \log t \quad (4)$$

The half-time of crystallization ($t_{1/2}$) is defined as the time required for relative crystallinity up to 50%, which can be calculated by the following equation.

$$t_{1/2} = \left(\frac{\ln 2}{K} \right)^{1/n} \quad (5)$$

Polarized light microscopy (PLM) with Nikon's Eclipse E600W was performed to observe the isothermal crystallization process of samples. A Linkam CFI60 hot stage was used to control the temperature with an accuracy of ± 0.1 °C. The samples were first clamped with a glass sheet and placed in a hot stage to melt at 200 °C for 5 min and then cooled to the desired crystallization temperature at 60 °C/min for the isothermal crystallization observation.

Wide-angle X-ray diffraction (WAXD) analysis was conducted with a Philips Pert Pro diffractometer with Cu K α radiation ($\lambda = 1.54$ Å). The process of the test was under the protection of nitrogen, and the scanning 2θ range was 5–40°. Meanwhile, small-angle X-ray scattering (SAXS) measurement was performed using a semiconductor pixel detector (Pilatus 200 K) with the Cu K α radiation ($\lambda = 1.54$ Å), a Nano-inXider vertical SAXS system was operated at 50 kV and 0.6 mA.

The melt flow index (MFI) of the samples was measured by using an QT-400B type equipment at 230 °C with a load of 2.16 kg, following the ASTM D1238 standard.

RESULTS

St Content in PP-g-PS. The FTIR spectra of iPP and PP-g-PS are plotted in Figure 1. Compared with the infrared spectrum

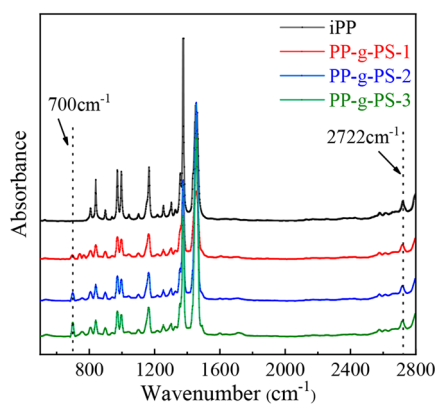


Figure 1. FTIR spectra of iPP and PP-g-PS.

of iPP, a new peak corresponding to the stretching of the hydrogen atoms of the benzene ring appears at 700 cm^{-1} for PP-g-PS, indicative of the successful grafting of St onto the iPP chain. Generally, the peak at 2722 cm^{-1} corresponding to the iPP backbone could be chosen as the internal reference,³² the peak at 700 cm^{-1} corresponding to St can be used to determine the content of St grafting segments. For that purpose, a series of mixtures of iPP and PS with different quantitative proportions were prepared to establish a standard curve of FTIR absorption ratio at 700 and 2722 cm^{-1} (A_{700}/A_{2722}) versus PS concentration in PS/iPP blends. As shown in Figure 2, the standard curve exhibits a linear relationship. On the basis of the standard curve, the grafting content of St in PP-g-PS-1, PP-g-PS-2, and PP-g-PS-3 can be calculated as 1.1, 2.0, and 3.4 wt %, respectively.

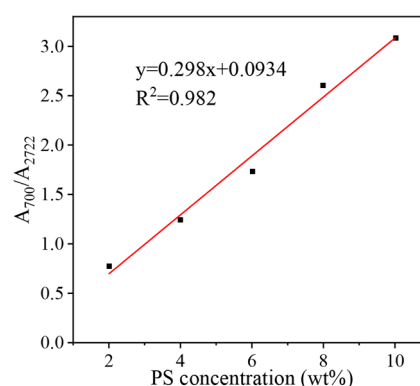


Figure 2. Curve of A_{700}/A_{2722} versus PS concentration in iPP/PS blends.

Crystallization and Melting Behaviors of PP-g-PS. The nonisothermal crystallization and melting behaviors of iPP and PP-g-PS are shown in Figure 3. The corresponding parameters of crystallization and melting are listed in Table 2. It is expected that the introduction of St grafting segments results in a slight decrease in crystallinity due to the increase of polymer chain irregularity, and a decrease of the melting temperature because of the addition of amorphous polymer to crystalline polymer as Deng's report.³³ Interestingly, the introduction of St grafting segments can effectively promote the crystallization of iPP. More specifically, when 1.1 wt % St is grafted onto the iPP chain, a significant elevation in the crystallization temperature is observed from 110.1 to 116.8 °C; when the content of St grafting segments further increases to 3.4 wt %, the crystallization temperature reaches 120.5 °C.

The isothermal crystallization curves at 130 °C have also illustrated that the introduction of St grafting segments can remarkably promote the crystallization of iPP. As shown in Figure 4, the time required for complete isothermal crystallization for PP-g-PS is obviously faster than that of iPP. Moreover, the higher the St content, the faster the isothermal crystallization of PP-g-PS.

The isothermal crystallization thermograms of iPP and PP-g-PS-3 at different crystallization temperatures are compared in Figure 5. Apparently, the required time to achieve relatively comprehensive crystallinity increases with the increase of isothermal crystallization temperature, which implies a decrease in the rate of crystallization. According to Avrami equation, the K value of PP-g-PS-3 is $1.2 \times 10^{-2} \text{ min}^{-1}$ whereas that of iPP is $42 \times 10^{-2} \text{ min}^{-1}$ at 132 °C. This further confirms that the introduction of St grafting segments can significantly accelerate the crystallization of iPP. Moreover, it should be noted that the n values of iPP and PP-g-PS-3 are almost the same, which infers that the introduction of St grafting segments does not change the mechanism of nucleation and crystal growth of iPP.

The isothermal crystallization behaviors of iPP and PP-g-PS were further online observed by PLM at 140 °C, as shown in Figure 6. Compared with iPP, PP-g-PS could form much more and finer crystalline grains under the same isothermal conditions. Moreover, this phenomenon is more remarkable as the increase of St content. These results further confirm that the introduction of St grafting segments into iPP could effectively promote the formation of iPP crystalline grains.

To quantify the growth rate of spherulites from the PLM images, the spherulite size was measured using custom graphics software (ImageJ pro) at each time interval, as shown in Figure

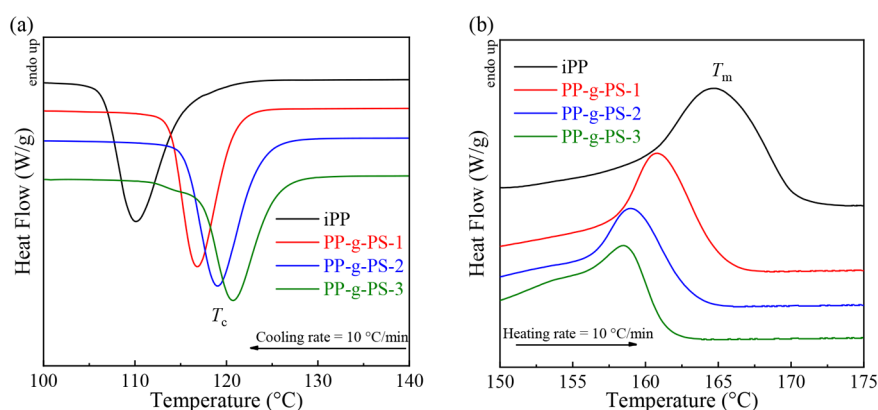


Figure 3. Nonisothermal crystallization curves (a) and melting curves (b) of iPP and PP-g-PS.

Table 2. Characteristics of iPP and PP-g-PS

polymer	St content (wt %)	M_w (kDa)	PDI	T_c (°C)	T_m (°C)	ϕ_m (%)	ρ_f^a (g/cm ³)	ρ_a^b (g/cm ³)
iPP	0	476	4.24	110.1	164.7	45.5	0.901	0.863
PP-g-PS-1	1.1	148	3.26	116.8	161.4	44.6	0.906	0.874
PP-g-PS-2	2.0	178	4.15	119.1	160.5	42.1	0.905	0.875
PP-g-PS-3	3.4	305	4.31	120.5	158.5	40.7	0.912	0.889

^aFrom drainage method at room temperature according to ASTM D792 standard. ^bUsing $\rho_a = (\rho_f - \rho_c \phi_m) / (1 - \phi_m)$.

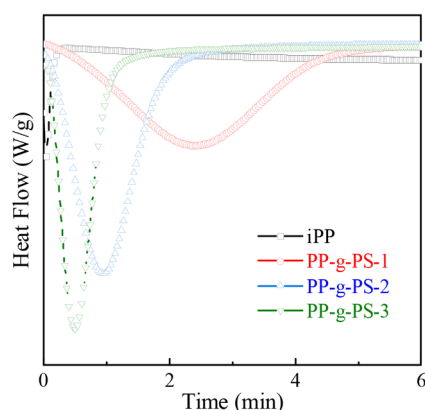


Figure 4. Isothermal crystallization curves of iPP and PP-g-PS at 130 °C.

7. Generally, the single crystal average size increased with time until the melt-spherulite interfaces approach one another. However, when the crystallization process initially goes, unimpinged interfaces will grow at the same rate without the interfere from adjacent spherulites. Obviously, PP-g-PS performed a relatively slower growth rate than that of iPP. Moreover, as the PS content increasing, the growth rate of single crystal size decreased significantly. Therefore, PS grafting segments would probably limit the growth of single crystal but promote the process of nucleation, i.e., formation of more crystalline grains, which in turn shows an accelerated rate in a macroscopic sense.

Figure 8 presents the WAXD patterns of iPP and PP-g-PS. The diffraction peak positions of iPP are concentrated at 14.0°, 16.8°, 18.6°, 21.1°, and 21.8°, which correspond respectively to the (110), (040), (130), (111), and (041) reflections of the traditional α -crystal. Interestingly, there is a typical characteristic peak at 20.1° for PP-g-PS, which corresponds to the (117) reflection of γ -crystal.³⁴ Moreover, the peak area at 20.1°

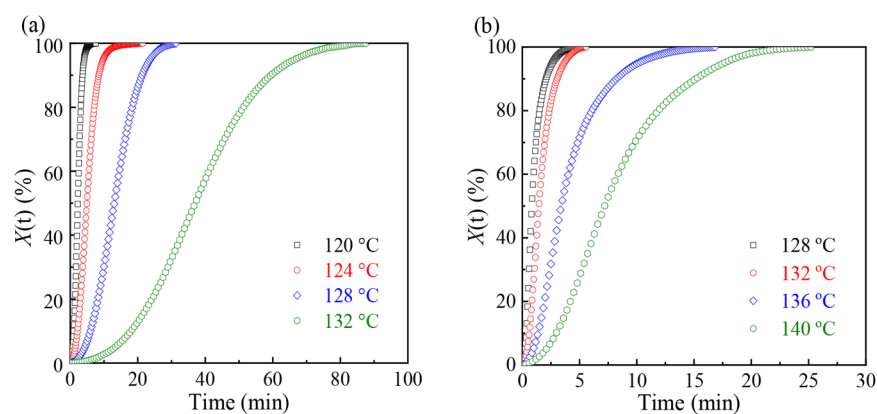


Figure 5. Development of relative crystallinities of iPP (a) and PP-g-PS-3 (b) as a function of time at different temperatures during isothermal crystallization.

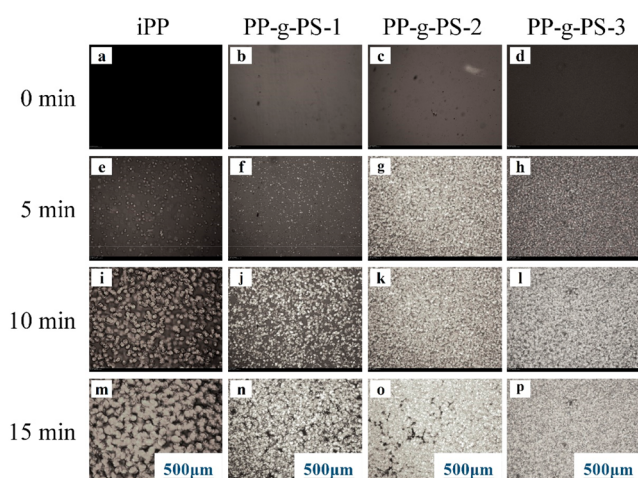


Figure 6. Isothermal crystallization images observed by PLM at different times. iPP: (a), (e), (i), (m). PP-g-PS-1: (b), (f), (j), (n). PP-g-PS-2: (c), (g), (k), (o). PP-g-PS-3: (d), (h), (l), (p).

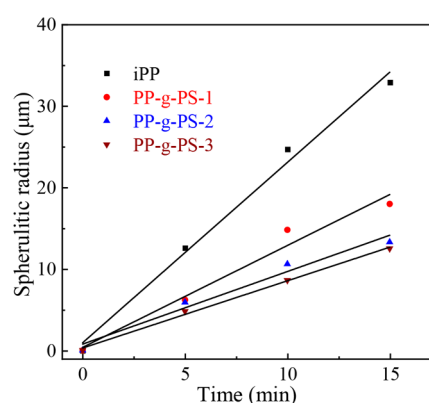


Figure 7. Spherulitic radius of iPP and PP-g-PS as a function of time at 140 °C.

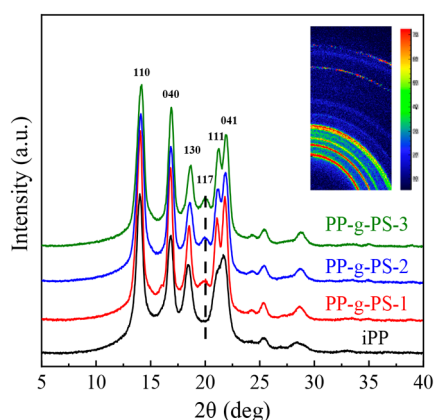


Figure 8. WAXD patterns of iPP and PP-g-PS.

progressively increases with the increase of St content in PP-g-PS. As reported,³⁵ the γ -crystal of iPP results mainly from high pressure or the existence of chain defects. Apparently, the appearance of γ -crystal in PP-g-PS might be ascribed to the produced stereo- and regio-defects because of the introduction of St grafting segments.

DISCUSSION

Effect of Degradation on the Crystallization. During the melt grafting of iPP, iPP chains tend to degrade partly to

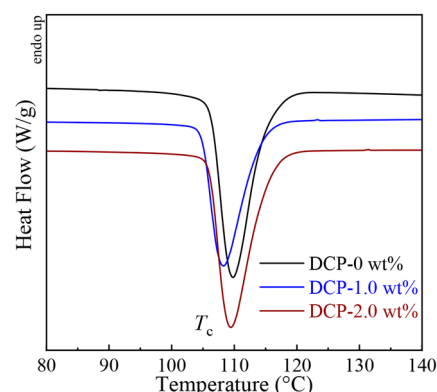


Figure 9. Nonisothermal crystallization behavior of iPP and degraded iPP.

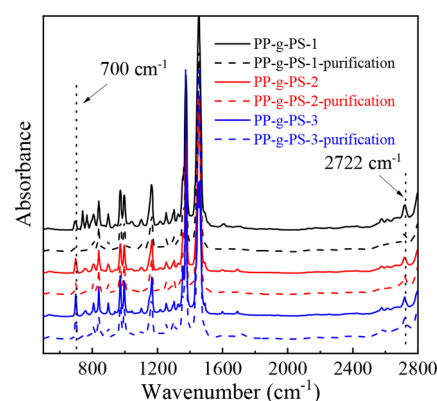


Figure 10. FTIR spectra of PP-g-PS-1,2,3 before and after purification.

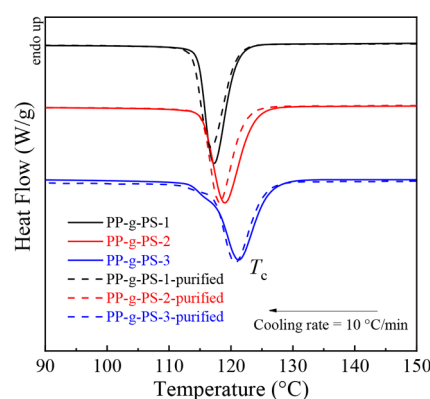


Figure 11. Nonisothermal crystallization curves of PP-g-PS-1,2,3 before and after purification.

produce the modified iPP with lower molecular weight (higher MFI) due to the β -scission. As summarized in Table 2, with the introduction of St into the iPP matrix, the M_w value decreases from 476 000 to a range between 148 000 and 305 000, depending on the St content. Meanwhile, the corresponding MFI values of PP-g-PS ranging from 27.9 to 41.8 g/10 min are also higher than that of iPP. As the previous literature reported,³⁶ the decrease in the molecular weight of iPP tends

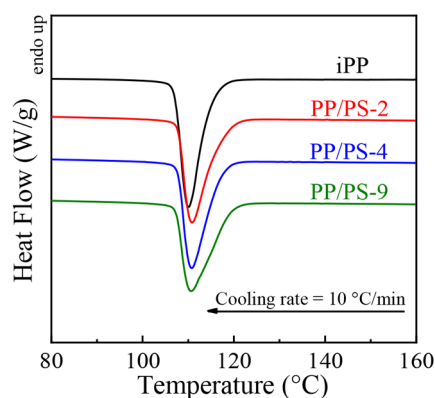


Figure 12. Nonisothermal crystallization curves of iPP and PP/PS blends with a cooling rate of 10 °C/min.

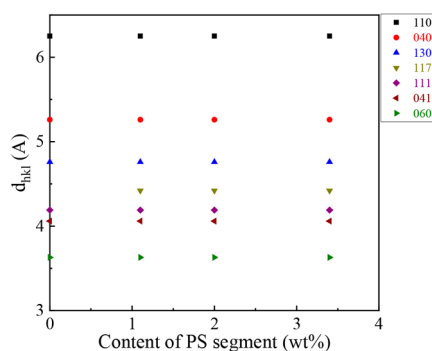


Figure 13. Lattice spacings of crystal of iPP and PP-g-PS.

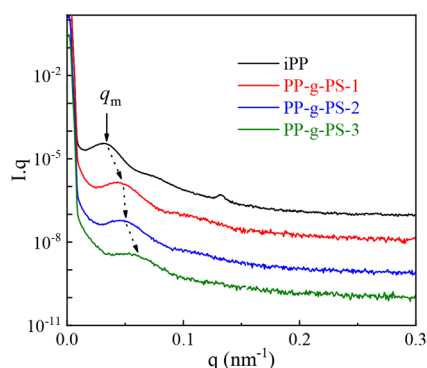


Figure 14. SAXS patterns of iPP and PP-g-PS.

Table 3. Average Crystal Dimensions of iPP and PP-g-PS

polymer	ϕ_v (%)	L (nm)	l_c (nm)	l_a (nm)
iPP	43.2	19.0	8.2	10.8
PP-g-PS-1	42.6	13.7	5.8	7.8
PP-g-PS-2	40.2	12.8	5.2	7.7
PP-g-PS-3	39.2	10.8	4.2	6.6

to accelerate the crystallization and increase the spherulite growth rate. Thus, the influence of the produced low molecular weight iPP on the crystallization during the melt grafting process may not be ignored. However, it should be pointed out that grafting St could inhibit the degradation. With more St grafted to iPP, the increase of M_w and the decrease of MFI values does not inhibit the crystallization but enhances the crystallization temperature to above 120 °C, which indicates that the

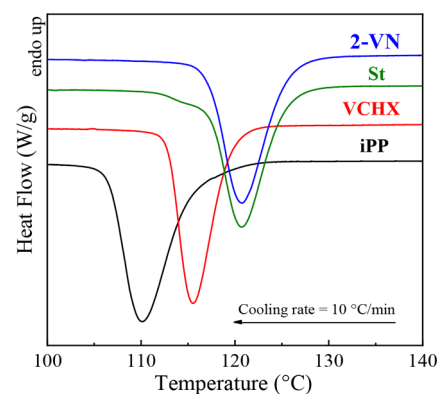


Figure 15. Nonisothermal crystallization curves of modified iPP with or without the phenyl groups.

degradation may be not the main reason for the crystallization improvement of modified PP by grafting St. In order to further confirm this deduction, the DCP without St was blended into iPP at 190 °C for 10 min to obtain the degraded iPPs. When the dosage of DCP is 1 and 2 wt %, which are the same as that for the synthesis of PP-g-PS-2 and PP-g-PS-3, the MFI of iPP increases from 14.4 to 444 and 876 g/10 min, respectively. The crystallization curves of those degraded iPP samples were presented in Figure 9. It can be seen that the severe degradation of iPP has a slight impact on the crystallization temperature, which verifies the degradation of iPP is not the main factor for the enhancement of crystallization temperature of modified PP by grafting St.

Effect of PS Nucleation on the Crystallization. It is uncertain whether the nucleation of nongrafted PS could promote the crystallization behavior. Although these grafted samples have been purified to remove the nongrafted PS and nonreacted monomers, it is essential to verify the effectiveness of purification process. To specify, the FIIR spectra of the PP-g-PS samples before and after purification are compared in Figure 10. On the basis of the calculated values of A_{700}/A_{2722} , the St contents in PP-g-PS-1,2,3 samples are 1.4, 3.5, and 4.1 wt % respectively. After purification, they reduce to 1.1, 2.0, and 3.4 wt %. Thus, the process of purification could significantly remove the nongrafted PS. Additionally, the nonisothermal crystallization curves of PP-g-PS before and after purification are shown in Figure 11. Apparently, the crystallization temperature almost remains unchanged, which further demonstrates that the nongrafted PS would hardly promote the crystallization of iPP.

For comparing the effect between reactive and physical introduction of St on the crystallization behavior of PP-g-PS, the iPP and PS blends with different proportions were prepared for studying their nonisothermal crystallization behavior. As shown in Figure 12, the addition of 2, 4, and 9 wt % PS, denoted respectively as PP/PS-2, PP/PS-4, and PP/PS-9, has little effect on the crystallization temperature of iPP, indicating that St should be only grafted onto the main chain of iPP to promote the crystallization of iPP.

Analysis of Lamellae Crystal Characteristics of PP-g-PS. Understanding the role of St grafting segments on the crystallization process of modified iPP is essential, yet inadequate. Because PS is an amorphous polymer, the traditional crystallization mechanism, i.e., “dimensional lattice matching theory”, is unable to explain the role of St grafting segments. The above discussion has illustrated that the PS nucleation and changes of molecular weight cannot modify the

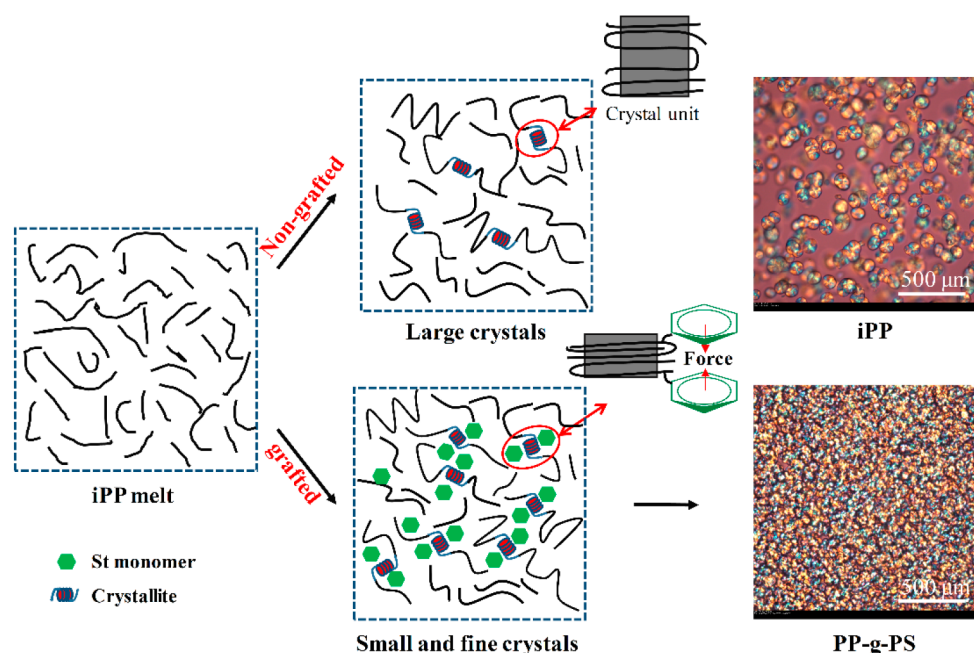


Figure 16. Diagrammatic drawing of crystallization processes of iPP and PP-g-PS.

crystallization behavior effectively and sufficiently. Therefore, it may be ascribed to the interaction between the PS side chain and iPP backbones. Likewise, GÜldoğ̃a et al.³⁷ reported the chain interaction of grafted PP, such as polarity, indeed resulted in the different crystallization. Analysis of crystal structure helps locate the function of St grafting segments on the crystal units of PP-g-PS, in turn to reveal the relationship between iPP backbones and St grafting chain.

On the basis of Bragg's law, the lattice spacings (d_{hkl}) of crystal before and after grafting St are calculated by following equation:

$$n\lambda = 2d[\sin \theta] \quad (6)$$

where n is an integer, λ is the wavelength of the radiation that equals 0.1542 nm, and θ is half of the diffraction angle. From Figure 13, it can be known that the St grafting segments have no effect on the lattice spacing, which illustrated that the unit cell size of the iPP crystal is unaffected by the introduction of St grafting segments. In other words, the PS grafting segments are excluded from the crystal and exist in the amorphous region, while the regular arrangement of iPP chains constitutes the crystal units.

Meanwhile, Figure 14 presents the SAXS patterns of iPP and PP-g-PS. A well-defined interference peak illustrates the existence of crystalline lamellae structure. The simple equation, $L = 2\pi/q_m$, could effectively calculate the thickness in the direction of the long axis, as listed in Table 3. The period crystalline phase consists of stacks of alternating crystalline and amorphous lamellae, i.e., $L = l_c + l_a$, where l_c and l_a represent the thickness of the crystal lamella and the amorphous fold, respectively. The values of $l_c = \phi_v L$ and $l_a = (L - l_c)$ could be acquired from the volume fraction (ϕ_v) of crystal phase expressed by the following equation:

$$\phi_v = \frac{\phi_m \rho_a}{\phi_m \rho_a + (1 - \phi_m) \rho_c} \quad (7)$$

where ρ_c is the density of iPP with 100% crystal (0.946 g/cm³)³⁸ and ρ_a is the density of amorphous phase, which can be calculated from eq 8 because St grafting segments are located

exclusively in the amorphous phase do not affect the density of crystalline phase.

$$\rho_t = \phi_m \rho_c + (1 - \phi_m) \rho_a \quad (8)$$

The bulk densities (ρ_t) of iPP and PP-g-PS were measured by the drainage method, as listed in Table 2. Thus, the values of l_c and l_a were acquired, as listed in Table 3. The values of L , l_c , and l_a remarkably decrease almost 30–50% upon the insertion of St grafting segments. Moreover, the higher the St content, the shorter the length of a period lamellae. This indicates that the St grafting segments locating in the amorphous region can hinder the growth of unit cells and the development of structure lamellae.

Effect of Graft Monomers on the Crystallization.

Structural analysis of grafted monomers would help establish, at least in broad outlines, the relationship between the crystallization behavior of modified iPP and sorts of monomers. In the field of polymeric NAs, their structural characteristics reveals that they all have phenyl groups, which may act as a growth surface for the crystal and induce the helix chains of iPP chain to epitaxially crystallize on it.⁸ As reported, the π – π conjugation from the adjacent phenyl rings helps stabilize the crystals and promote the crystallization.^{39,40} In terms of melt grafting reaction, do the monomers without the phenyl groups have a similar effect of St to improve the crystallization of iPP? VCHX and 2-VB were grafted onto iPP whose molar dosages were the same as that of St for the preparation of PP-g-PS-3. Unlike St, the six-membered ring in the VCHX molecule is the hexane ring rather than the benzene ring. Although the grafted PP by VCHX can also improve the crystallization temperature of iPP, the degree of improvement is much lower than that of the grafted PP by St, as shown in Figure 15. However, when 2-VB with the biphenyl ring is grafted into iPP, the crystallization temperature can increase significantly, which is similar to that of the grafted PP by St. Thus, the presence of a benzene ring conjugated π – π interaction with the helix chains of iPP may be of vital importance in promoting the crystallization of iPP.

Description for the Crystallization Route of PP-g-PS.

On the basis of the above discussions, a possible diagrammatic mechanism shown in Figure 16 is proposed to specifically explain the interesting crystallization behavior of PP-g-PS. For iPP, the crystallization route can be described as follows: first, a large amount of chain sequences starts to stack from the melt at a large supercooling degree, which forms an ordered structure, i.e., crystal nucleus. Subsequently, the crystal nuclei are stacked into macroscopic spherulites, as shown in the PLM image of iPP. When St is introduced into the iPP backbone, the main chain is divided into many isotactic sequences by the St grafting segments. The rigid structure of benzene ring easily squeezes the PP chain segments to form the relatively ordered structures. Moreover, the mutual π - π conjugation between the benzene rings and the helix chains of iPP also can stabilize the structure. As a result, the crystal nuclei are easier to form. As the branches could retard the mobility and reptation of iPP chains,^{41,42} the St grafting segments located in the amorphous phase can hinder the growth of crystals, which leads to finer crystals. Thus, the introduction of St grafting segments could effectively promote the crystallization behavior (e.g., crystallization temperature, crystallization rate), but restrain the mobility of iPP backbones and limit the growth of single crystals.

CONCLUSIONS

A modified PP was prepared by the melt grafting of St onto iPP. The insertion of St grafting segments into iPP could extremely accelerate iPP crystallization, generate finer and much more crystals, and enhance remarkably the crystallization temperature. Moreover, a new γ -crystal appeared. In order to investigate the underlying mechanism, the influences of the existence of iPP with low molecular weight and nongrafting PS on the crystallization behavior of iPP were investigated. Results showed only when the PS chains were grafted onto the iPP chain could they boost the crystallization effectively. The rigid structure of benzene rings in St grafting segments can squeeze easily iPP chains to form the relatively ordered structures. Simultaneously, the mutual π - π conjugation between the benzene rings and the helix chains of iPP can further stabilize the ordered structure and promote the crystallization behavior. Moreover, the St grafting segments located in the amorphous phase can hinder the growth of crystals, leading to finer and more crystals.

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Notes

The authors declare no competing financial interest.

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